

Scientists See Tiny Particles Make Ripples in a Super-Hot 'Soup'

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Right after the **universe** began, it was trillions of degrees hot. Tiny bits called **quarks** and gluons zoomed around in a hot "soup" called quark-gluon **plasma**. It lasted only a few millionths of a second.

Today, scientists can make tiny drops of this soup at CERN's Large Hadron Collider in Switzerland. They smash heavy atoms, like lead, together very fast.

Now a team led by **physicists** from MIT has found that **quarks** leave **wakes** as they move through the **plasma**. It is like a duck making ripples on a pond. This shows the **plasma** acts like a liquid, with waves, splashes and swirls.

"Now we see the **plasma** is incredibly **dense**," said MIT physicist Yen-Jie Lee. He said it can even slow down a quark.

The team used a new way to spot the **wakes**. Measuring them may help scientists learn about the **universe's** first moments.

Word Treasure

universe -- Everything that exists, including all of space, stars and planets.

quarks -- Super tiny bits that make up the inside of atoms.

plasma -- A super-hot form of matter that is not a solid, liquid or regular gas.

wakes -- Trails of ripples left behind by something moving through water.

physicists -- Scientists who study matter, energy and how things move.

dense -- Packed very tightly together.

Background you need

Everything around us is made of atoms, and atoms are made of even smaller bits like quarks.

When the universe was brand new, it was so hot that these tiny bits could not stick together. Scientists use giant machines to smash atoms together so they can study what the early universe was like.

Break it down

WHO: A team led by MIT physicists, including Yen-Jie Lee

WHAT: They saw quarks leave ripples in super-hot quark-gluon plasma

WHERE: CERN's Large Hadron Collider in Switzerland

WHY: To learn what the universe was like in its very first moments

Why it matters

It helps us understand how the stuff that makes up you and everything around you first formed.

Test what you remember

1. How hot was the universe right after it began?
 - ☐ (A) Room temperature
 - ☐ (B) 100 degrees
 - ☐ (C) Freezing cold
 - ☐ (D) Trillions of degrees
2. What is the hot 'soup' called?
 - ☐ (A) Star dust
 - ☐ (B) Quark-gluon plasma
 - ☐ (C) Lava
 - ☐ (D) Ocean water
3. Where do scientists make tiny drops of it today?
 - ☐ (A) On the Moon
 - ☐ (B) In a kitchen
 - ☐ (C) At CERN's Large Hadron Collider in Switzerland
 - ☐ (D) In a volcano
4. What do quarks leave as they move through the plasma?
 - ☐ (A) Nothing
 - ☐ (B) Wakes, like a duck on a pond
 - ☐ (C) Smoke
 - ☐ (D) Footprints
5. What does this show the plasma acts like?
 - ☐ (A) A liquid
 - ☐ (B) A balloon
 - ☐ (C) A magnet
 - ☐ (D) A rock
6. Who led the team?
 - ☐ (A) Chefs
 - ☐ (B) Doctors from a hospital
 - ☐ (C) Physicists from MIT
 - ☐ (D) Astronauts

Answer key (try the quiz first!): 1: D 2: B 3: C 4: B 5: A 6: C

1. Yen-Jie Lee: Says the plasma is incredibly dense and can even slow down a quark.